RL Control for QEC Under Correlated Noise

Steane-Code Simulation + Backend/Training Pipeline

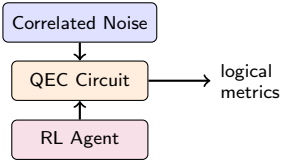
Wenzheng Dong

Physics Seminar

March 11, 2026

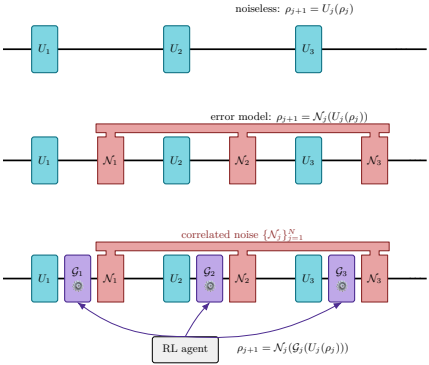
Motivation

- Realistic hardware noise is often **temporally-correlated**, with various **types/origins** (drift, low-frequency fluctuations, crosstalk-like effects).
- Ignoring this can overestimate **QEC** performance under idealized i.i.d. assumptions.
- Goal: quantify how **reinforcement learning (RL)** control improves logical performance vs fixed baseline.
- Practical challenge: end-to-end throughput under **composed** correlated channels.



AI teaser

System Picture



Noisy controlled evolution:

$$\rho_{j+1} = \mathcal{N}_j^{\mathrm{corr}}\Big(\mathcal{G}_j^{(u_j)}\big(U_j \rho_j U_j^{\dagger}\big)\Big).$$

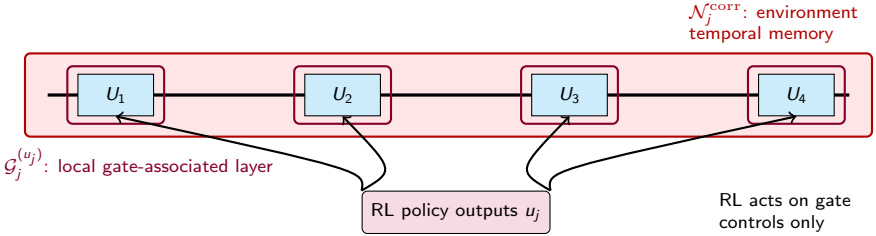
Physical Meaning of $\mathcal{G}$ and $\mathcal{N}$

- **Yes**: gate-depolarization is used as an effective model of **gate-operation-associated** noise.
- **Yes**: correlated channel is used as **environment/background** noise with temporal memory.
- $\mathcal{G}_j^{(u_j)}$ acts on gate controls (1q/2q slots); in the composed model, $\mathcal{N}_j^{\mathrm{corr}}$ strength can couple to control mismatch, while correlation structure (f, γ) remains environmental.

$$\rho_{j+1} = \mathcal{N}_j^{\mathrm{corr}}\Big(\mathcal{G}_j^{(u_j)}\big(U_j \rho_j U_j^\dagger\big)\Big).$$

Here u_j is produced by RL policy; $\mathcal{N}_j^{\mathrm{corr}}$ mainly acts during idle windows.

Physical Picture of Two Noise Layers



$$\rho_{j+1} = \mathcal{N}_j^{\mathrm{corr}}\Big(\mathcal{G}_j^{(u_j)}\big(U_j \rho_j U_j^\dagger\big)\Big).$$

Google-Style Noise Coverage: What We Do Not Yet Model

Reported in Google paper / supplement

- Pauli characterization includes 1q, 2q, measurement, and DD-during-readout/reset error components.
- Drift scenarios include step-like, sinusoidal, and stroboscopic profiles over CZ coupling, XY amplitude, and frequency.
- Control space includes pulse-level knobs: DRAG terms, CZ detunings, transfer-function settling/reflection, virtual-Z phase corrections.

Source anchors: Google paper Fig. 4 text; Supplement Sec. I.A, II.A, V.A/B.

Not yet explicit in our current simulator

- No explicit measurement/readout/reset error channel (we focus on gate depolarization + correlated idle Pauli).
- No explicit pulse-transfer-function distortion model.
- No explicit context-dependent coherent phase/crosstalk error model.
- Control-to-noise map is low-dimensional effective mapping, not full pulse-parameter space per gate.

Important Noise Directions Beyond Both Studies

- **Leakage dynamics**: explicit $|2\rangle$-state population flow across cycles, including leakage propagation and repumping.
- **Non-Pauli coherent accumulation**: systematic over/under-rotation, spectator-ZZ, and coherent phase buildup not reduced to depolarizing/Pauli.
- **Burst/heavy-tail events**: rare but high-impact events (e.g., quasiparticle bursts / cosmic-ray-like faults) causing multi-qubit transients.
- **Multi-timescale nonstationarity**: $1/f$ + random-walk + abrupt jumps, beyond simple sinusoidal drift or two-state Markov telegraph.
- **Long-range spatial correlation**: chip-wide or bus-mediated correlated faults (including correlated readout excursions).

These directions likely require a richer simulator architecture (hybrid hidden-state noise, pulse-level abstractions, and co-training with decoder adaptation).

Minimal Physics Model Used in Simulation

Control-to-gate-noise map (for gate event e, slot $s(e)$):

$$p_{1q}^{(e)} = \mathrm{clip}\Big(p_{1q}^{\mathrm{base}} + \alpha_{1q}\big(u_{1q,s(e)} - u_{1q,s(e)}^{\star}(t)\big)^2,\ 0,\ p_{\max}\Big)\,,$$

$$p_{2q}^{(e)} = \mathrm{clip}\Big(p_{2q}^{\mathrm{base}} + \alpha_{2q}\big(u_{2q,s(e)} - u_{2q,s(e)}^{\star}(t)\big)^2,\ 0,\ p_{\max}\Big)\,.$$

Correlated environmental channel (two-state Markov telegraph):

$$\gamma = \frac{1 - e^{-f\Delta t}}{2}, \qquad T = \begin{bmatrix} 1-\gamma & \gamma \\ \gamma & 1-\gamma \end{bmatrix}, \qquad \Delta t = 200\,\mathrm{ns}.$$

$$p_{\mathrm{target}} = \mathrm{clip}\Big(\big(p_{1q}^{\mathrm{base}} + \alpha_{1q}\overline{(u-u^{\star})^2}\big)g, 0, p_{\max}\Big)\,, \qquad p_{\mathrm{idle}} = 1 - (1 - p_{\mathrm{target}})^{1/N_w}.$$

Experimental Setup (Physics View)

- Platform-level model: Steane memory experiment with repeated stabilizer rounds.
- Noise decomposition: gate-associated effective depolarization + temporally correlated environmental idle noise.
- Control objective: RL learns gate controls to improve logical metrics under non-i.i.d. noise.
- Main comparison: learned policy vs fixed-zero policy over many seeds.

What We Optimized Practically

- Simulator kernels for correlated/composed channels (to make large-seed studies feasible).
- RL-side training/evaluation throughput for stable statistical conclusions.
- Staged benchmark design over (f, g), then confirm runs with increasing seed count.

Backend Speedup (Kernel-Dominated A/B)

Channel	old (s)	new (s)	speedup
`google_gate_specific`	0.284	0.311	0.91x
`correlated_pauli_noise_channel`	4.759	0.437	10.89x
`composed_google_gate_specific_correlated`	10.670	0.597	17.87x

For target correlated/composed channels, simulator-side acceleration is substantial.

Benchmark Plan for RL Advantage

- Grid and focus over correlated parameters:

$$f \in \{10^2, 10^3, 10^4\}, \quad g \in \{0.4, 1.0, 1.6\}.$$

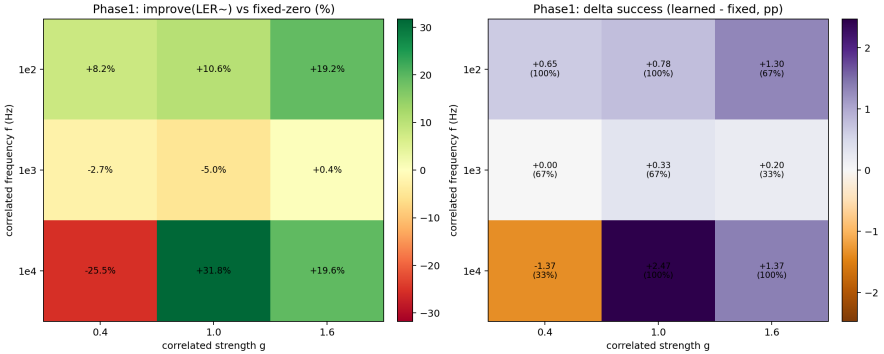
- Multi-phase design:
  - quick grid $\rightarrow$ pilot focus $\rightarrow$ confirm (10 seeds) $\rightarrow$ confirm (40 seeds).
- Key condition selected by stability and effect size:

$$f = 10^4, \; g = 0.4.$$

Fig.13: Phase1 Grid (What This Figure Means)

Definitions:

- **Phase1** = quick screen on 9 (f, g) points (3 frequencies $\times$ 3 strengths).
- **Δsuccess** = `success_rate(learned)` $-$ `success_rate(fixed_zero)`.
- **learned-fixed (pp)** means this difference in **percentage points** (pp).

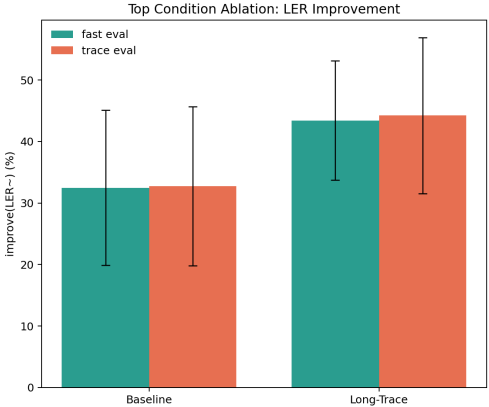


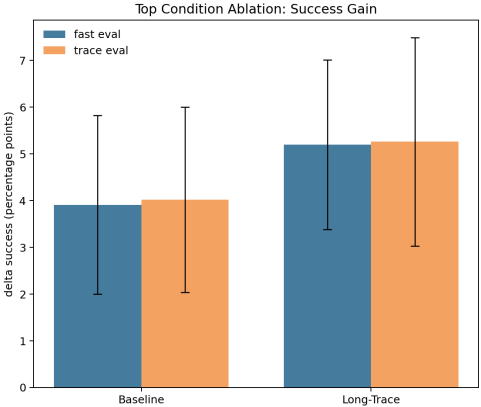
This is a screening map to select conditions for deeper confirm runs.

Fig.14: Top-Condition Ablation (What Is Being Compared)

Ablation = controlled comparison where only one design choice changes at a time (same top condition $f = 10^4, g = 0.4$, same benchmark protocol).

- **Fast eval**: no shot traces; uses summary/proxy metrics; much faster.
- **Trace eval**: stores shot traces and detector histories; higher fidelity, slower.

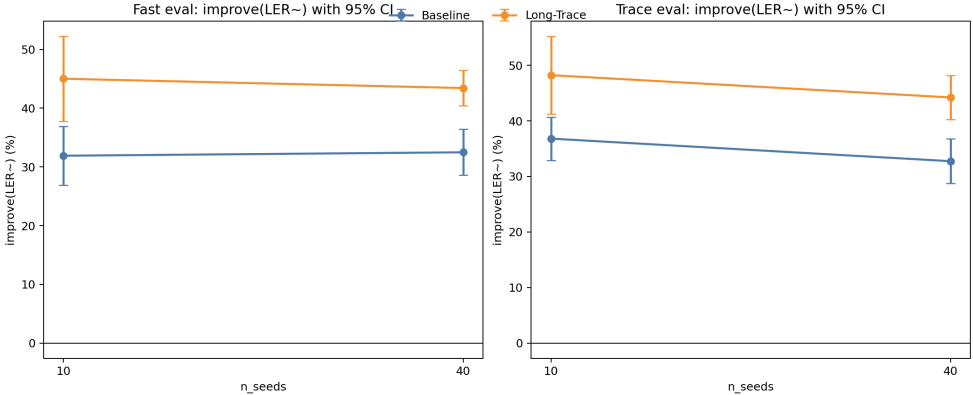




Error bars here are `std` (not 95% CI).

Fig.15: Confirm10 vs Confirm40 (Why Two Confirms)

- **Confirm10** = same condition and model, averaged over 10 random seeds.
- **Confirm40** = same thing, but 40 seeds for tighter uncertainty.
- Purpose: not to change method, only to improve statistical confidence.



CI half-width formula: $\mathrm{CI95}_{1/2} = 1.96 \cdot \mathrm{std}/\sqrt{n_{\text{seeds}}}$.

Numerical Result at $f = 10^4,\ g = 0.4$

Metric shown: $\mathrm{improve}(\mathrm{LER} \sim)$ (mean $\pm$ CI95 half-width)

Run	Fast eval	Trace eval
confirm10 baseline	31.90% $\pm$ 5.02%	36.81% $\pm$ 3.88%
confirm10 long-trace	45.01% $\pm$ 7.23%	48.23% $\pm$ 7.01%
confirm40 baseline	32.49% $\pm$ 3.91%	32.74% $\pm$ 4.00%
confirm40 long-trace	43.43% $\pm$ 3.01%	44.22% $\pm$ 3.94%

Takeaways

- **RL advantage** is **positive** and **reproducible** under composed correlated noise.
- Increasing **seeds** ($10 \rightarrow 40$) narrows **uncertainty**; effect becomes more statistically stable.
- **Long-trace** variant consistently outperforms **baseline** on the top condition.
- **Backend optimization** is necessary but not sufficient; **training/eval protocol** matters.

Next Steps

- **Incorporate measurement errors**: add stabilizer readout channel (start with symmetric flip, then asymmetric/correlated variants).
- Extend controlled scans over broader (f, g) regimes with fixed evaluation protocol.
- Test transferability to additional correlated/noise families.
- Strengthen claims with pre-registered statistical criteria and standardized CI reporting.

Thank you.